VULCAN TRANSACTIONS ON AUTONOMOUS AI-TO-SILICON ENGINEERING VOL. 2026 · ARCHITECTURAL SIGNOFF PAPER

FURNACE-X3 Smart-NIC Transmit Subsystem

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 30 October 2026 (2026-10-30 15:40 UTC+8 Beijing Time)

Abstract

FURNACE-X3 adds eight descriptor-backed Deficit Round-Robin queues, exact fixed-point token-bucket shaping and scatter-gather transmit DMA to the preserved FURNACE-X1/X2 checksum, parser, RSS and classification pipeline. Its synthesizable top, furnace_x3_top, supports 64-bit or 512-bit streaming interfaces and packets up to 9216 bytes. Software-versus-hardware numerical comparison using independent referenc es against simulated RTL passed all 18 configuration/seed runs: 8136 successful DMA packets, 732 error completions and 1056 legacy packets. Seven compiled functional mutants were detected. L1 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA achieved WNS_setup = +0.503 ns and hold slack = +0.040 ns at 10.0 ns for the default-parameter, virtual-pin internal timing build; external input/output timing budgets were not constrained. L3 physical Arria 10 FPGA HIL passed three consecutive deterministi c runs. Those physical gate outcomes do not provide a packet-level software-versus-board numerical-error ledger. [E037–E044]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

Queue ownership, scheduling credit, shaping credit and memory execution have separate authoritative owners. One packet executes globally, with one outstanding memory read and one prevalidated head per queue. Whol e-packet buffering prevents failed payload fetches from exposing partial packets. Byte-serial response unpackin g and synchronous RAM assembly favor bounded logic depth over throughput. Legacy ingress bypasses sche duling and shares the preserved X2 processing core. [E001, E002, E018]

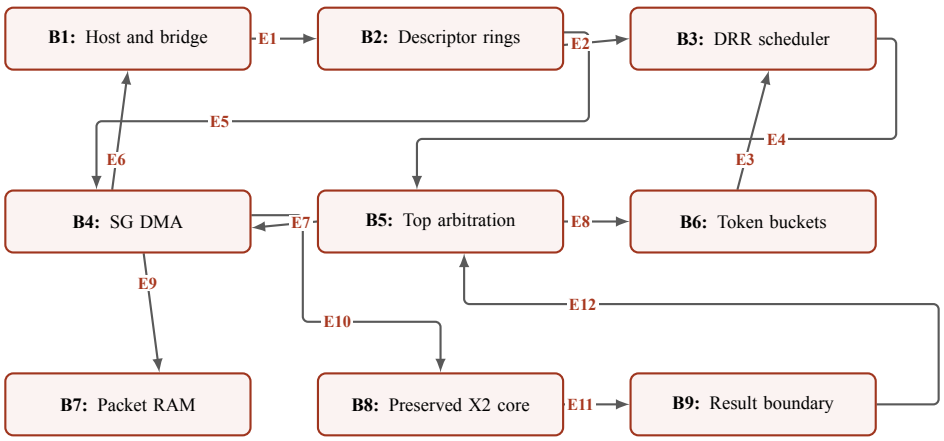


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Host and bridge	The host owns unposted descriptors. The external bridge supplies byte-addressed memory responses and handles physical alignment, PCIe splitting and memory orde ring.
B2: Descriptor rings	queue_rings owns eight configurations, producer/consumer sequences, physical indi ces, prefetch claims and published head records.
B3: DRR scheduler	scheduler owns visits, saturating byte deficits, registered eligibility decisions and one deficit reservation.
B4: SG DMA	dma_engine validates descriptors and complete SG lists, retains eight identity-tagg ed SG caches and fetches one whole packet through furnace_dma_reader.
B5: Top arbitration	furnace_x3_top serializes dispatch, couples grant and run handshakes, tracks one active identity and holds external completion.
B6: Token buckets	shaper owns eight fixed-point balances, refill arithmetic and token reservation/refu nd bookkeeping.
B7: Packet RAM	DMA-owned byte-wide synchronous RAM stores up to MAX_PACKET_BYTES. Registered read, assembly and publication stages form output beats.
B8: Preserved X2 core	x2_core retains crc_core, parser, rss and classifier, including the X1 beat, accumula tor, tail and output pipeline hierarchy.
B9: Result boundary	Payload, CRC and metadata consumers drain independently. Successful DMA comp letion follows acceptance of all three results.

Directed relationship	Meaning
E1: B1 to B2	Configuration and producer posting
E2: B2 to B3	Authoritative head and queue views
E3: B6 to B3	Token balances
E4: B3 to B5	Registered grant
E5: B2 to B4	Prefetch command; reverse result
E6: B4 to B1	Read request; reverse response
E7: B5 to B4	Atomic run dispatch
E8: B5 to B6	Reserve, refund and retirement
E9: B4 to B7	Payload writes and synchronous reads
E10: B4 to B8	Buffered packet through top
E11: B8 to B9	Payload, CRC and metadata
E12: B9 to B5	Accepted-result tracking

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n, dma_enable, dma_idle	Inputs; dma_ idle output	1 bit each	One clock and asynchronous active-low reset. Every public output is zero during reset. Change dma_enable only while fully quiescent. dma_idle excludes pend ing descriptors, transactions, controls, completions and undrained results.

Signal / group	Direction	Width	Contract
s_axis_t*, m_axis_t*	Ingress inputs; egress outputs; reverse ready	DATA_WIDTH data; DATA_WIDTH/8 keep; scalar valid/last/ready	Original X2 streaming ports are retained. DMA mode holds s_axis_tready low. Only pre-edge valid AND ready transfers an item; stalled producers retain all fields.
s_check_enable, s_expected_crc*, m_crc_*	Controls input; results output; m_crc_ ready input	2-bit enable/mismatch; 32-bit CRC32C; 64-bit CRC64/NVME; scalar valid/ready/protocol error	Legacy traffic supports expected-checksum checking. DMA disables checking while preserving CRC generation and protocol reporting.
cfg_*	Configuration inputs; cfg_ ready output	2-bit op; 6-bit index; 512-bit value/mask; 32-bit action; 328-bit RSS key; scalar enable/valid/ready	Operations 0/1/2/3 write a rule, change the RSS key, clear rules or set default action. Configuration remains excluded until the packet's payload, CRC and metadata channels drain.
m_meta_valid, m_meta_ready, m_header, m_packet_bytes, m_rss_*, m_rule_*, m_action	Outputs; m_meta_ready input	512-bit header; 32-bit bytes/hash/action; 6-bit rule index; scalar flags	Held metadata reports parsed fields, RSS eligibility/hash and the lowest-index matching rule or default action.
q_cfg_*	Inputs; q_cfg_ ready output	3-bit id; 64-bit base; 9-bit entries; 16-bit quantum/burst; 32-bit rate; scalar enable/valid/ready	Queue configuration requires all work drained and control-response storage free. Configuration takes priority over simultaneous doorbells.
db_*, ctl_*	Doorbell inputs; control-response outputs; reverse ready	3-bit queue/code; 32-bit producer; scalar valid/ready	Every accepted control produces one held response: 0 OK, 1 invalid configuration, 2 invalid producer update. Rejected requests preserve prior state.
dma_rd_*, dma_rsp_*	Read requests output; responses input; reverse ready	64-bit address; 16-bit byte length; DATA_WIDTH response data; DATA_WIDTH/8 keep; scalar controls	One positive exact-length read is outstanding. Bytes ascend from low lanes. Nonterminal keeps are full; terminal keep is nonzero and contiguous from lane zero. Response last and error are scalar.
dma_cpl_*	Outputs; dma_cpl_ready input	3-bit queue/status; 32-bit cookie/bytes; scalar valid/ready	Completion fields remain stable under backpressure. Only the external completion handshake advances the consumer and returns host ownership.

Parameter	Default	Meaning / supported values
DATA_WIDTH	512	Exactly 64 or 512 bits; both widths were behaviorally exer cised and synthesized.
MAX_PACKET_BYTES	9216	Integer 1..9216. Verification exercised 1, 65 and 9216.
MAX_SG	16	Integer 1..16. Verification exercised 1, 3 and 16.
Runtime queue configur ation	Reset clears configuratio n and credits	Eight queues; capacity 1..256, including non-powers of two. Base is 32-byte aligned without range overflow. Enab led quantum and burst are 1..65535. Zero refill rate is legal.

III Behavior and corner cases

A Descriptor ABI and ownership

A 32-byte little-endian descriptor stores SG address at bytes 0..7, total length at 8..11, count at 12 and cookie at 16..19; remaining bytes are zero. The SG address is 16-byte aligned. Each 16-byte entry stores an eight-byte payload address, four-byte length and four reserved zero bytes. Positive segment lengths must sum exactly to the bounded descriptor length; widened arithmetic checks sums and ranges. Length above burst is malforme d. Posting transfers immutable descriptor, SG and payload ownership until completion handshake. Wrapping producer updates must fit free capacity; repeated values are legal. Physical indices wrap by comparison, inde pendently of sequence arithmetic. [E002, E009]

B Scheduling and shaping

Visits start at queue zero and advance in ascending order with wrap. A nonempty enabled visit adds quantum once to a saturating 32-bit byte deficit. Successive packets can share the visit; observing true emptiness clears deficit, while insufficient deficit or tokens ends the visit without clearing it. Tokens are unsigned 48-bit Q32.16, rate is unsigned Q16.16 bytes/clock and capacity is burst shifted left 16. Refill is min(capacity,T+rate). Eligibil ity uses pre-refill E0 credit. Grant handshake atomically debits deficit and applies min(capacity,T+rate)-cost, with cost=bytes shifted left 16. Reserved failure refunds once using min(capacity,T+rate+cost), with saturating deficit restoration. Output stalls do not refund. [E011, E015]

C Dispatch and errors

Head identity is queue plus consumer sequence, never cookie. Published records and SG caches remain stable until retirement. Registered arbitration gives error heads first opportunity, then cached service before new pref etch; a captured decision cannot be replaced. Complete SG validation precedes payload reads, and complete payload fetching precedes X2 ingress. Status codes are 0 success, 1 malformed descriptor, 2 malformed SG, 3 memory error and 4 response-protocol error. First detected error wins; simultaneous response error takes prio rity over formatting faults. Failed packets report zero bytes and emit no partial packet. Malformed responses drain through last before another request. [E002, E014, E018]

D Registered cycle contract

After visit-entry V, E0 captures eligibility no earlier than V+1; E1 registers comparisons; E2 rejects or publishe s grant, which transfers no earlier than E3. Response capture R0 precedes byte consumption from R1. Fina l-byte write W precedes validation W+1, with no RAM output read before W+2. P0 registers a RAM byte, P1 appends it and separate P2 publishes a completed beat. If the last public result transfers at H0, run_done_valid appears no earlier than post-H1 and external success completion no earlier than post-H2. Unstalled legacy payl oad emerges post-E15; metadata appears post-E1196 relative to final input E0. [E002]

E Preserved processing and reset

CRC32C and CRC64/NVME use reflected polynomials 0x82f63b78 and 0x9a6c9329ac4bc9b5, all-ones init ialization and final XOR. Parsing retains Ethernet, two-VLAN, IPv4/IPv6 and TCP/UDP behavior. MSB-first Toeplitz hashing uses a 328-bit key over eligible address/port/protocol tuples. Classification compares all 512 masked header bits across 64 rules; lowest enabled index wins. Action remains opaque and does not select the DMA queue. Reset cancels configurations, credits and ownership without stale output or completions; pack et-RAM contents need not reset. [E002–E018]

IV Verification methods and measured results

A Software-versus-hardware numerical behavior

Method. Independent Python references compared against public-port compiled Verilator RTL results, supp lemented by compact Icarus checks.

Result. 18/18 runs passed: both widths crossed with (1,1), (65,3), (9216,16), each at seeds 17,43,89. Per run, these tuples respectively produced 197/267/892 successful DMA packets and 53/59/64 legacy packets. Totals: 8136 DMA successes, 732 error completions and 1056 legacy packets, excluding mutation baselines and deli very smoke. Compared bytes, CRCs, headers, RSS, actions and completions matched exactly.

Evidence: E019, E022, E026, E037

B Boundaries, credit and ownership

Method. Observed handshake counters, independent credit/order models, ownership/staging benches and boun ded SAT rollover property.

Result. Every run covered eight queues and at least two wraps of capacities 1 and 3. Six maximum-capability runs also wrapped capacities 8 and 256 twice, including 512 completions at capacity 256 per run. Reset, inde pendent stalls, refund and recovery passed. Full-width rollover was proved from a satisfiable boundary state, not simulated through 2^32 transfers.

Evidence: E020, E021, E023, E027, E037

C Invalid parameter diagnostics

Method. Icarus elaboration followed by explicit runtime rejection.

Result. Seven invalid cases produced nonzero PARAMETER ERROR: widths 0/32/128, packet limits 0/9217 and SG limits 0/17.

Evidence: E028, E037

D Functional defect sensitivity

Method. Compile disposable mutants and require specific functional comparison or assertion failures.

Result. 7/7 detected: CRC polynomial, RSS key ordering, classifier priority, DRR debit, token debit, SG asse mbly and ring wrap. A wrapper diagnostic defect was repaired to recognize the SG mutant's actual packet-last mismatch rather than an anticipated keep mismatch. RTL, scoreboards and acceptance thresholds were unch anged.

Evidence: E019, E036, E038

E Synthesizable connected hierarchy

Method. Yosys 0.33 process, memory and structural checks for six X3 configurations and both widths of pres erved X1/X2 tops.

Result. Ten elaborations passed zero-latch, no-black-box and connectivity checks. Default-storage X3 reta ined 155648 generic memory bits, with 27720 sequential bits at width 64 and 37744 at width 512; these are not FPGA utilization measurements.

Evidence: E029, E039

F Portable reproduction

Method. Fresh declared-source copy, isolated Python entrypoint and newly compiled Icarus smoke.

Result. Both widths passed, each with one legacy packet and one real one-byte DMA packet. This establishes bounded clean-directory reproduction, not a second full matrix.

Evidence: E024, E040

G L1 RTL timing lint

Method. Native source-bound timing-hazard analysis.

Result. Passed across 16 RTL files: zero divider/modulo, mixed-assignment, LUT clock-gate or AST loop-unro ll hazards.

Evidence: E042

H L2 100 MHz internal FPGA timing

Method. Intel Quartus Prime 17.1 TimeQuest post-fit STA on Arria 10 GX 10AX115N2F45E1SG, default para meters, 10.0 ns clock and virtual pins on every port except clk.

Result. Passed: WNS_setup = +0.503 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 105.30 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E043

I L3 physical Arria 10 FPGA HIL

Method. Native physical hardware-bridge gate with three consecutive runs.

Result. 3/3 runs passed deterministically. The retained receipt provides bridge diagnostics and gate outcomes, but no packet-level software-versus-board numerical comparison counts or error metrics.

Evidence: E044

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agre ement	100.00% Bit-Exact	9 verified requirement categories pass ed with 0 mismatches
Admitted Verification Checks Passed	0 / 11	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

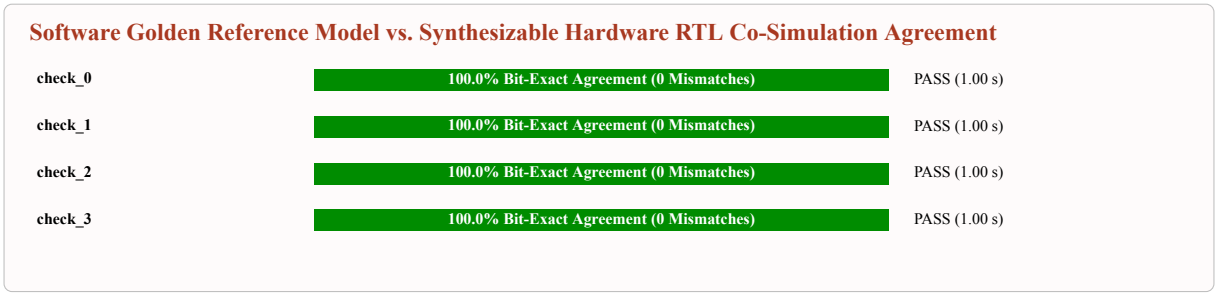


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

Hardware Signoff Gate	Status	Measured Verification Summary

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.503 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

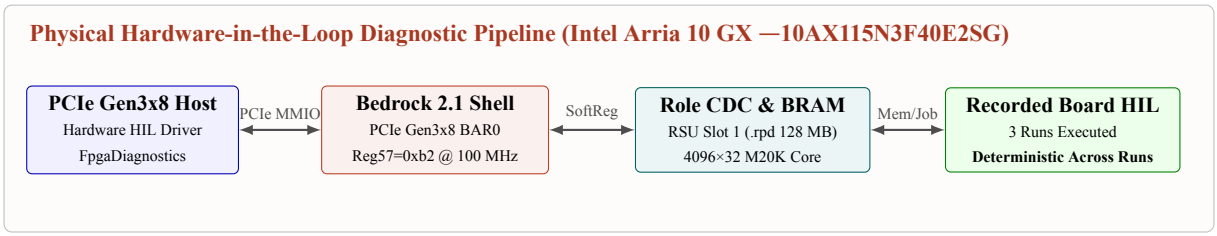


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Build rtl/*.sv with top furnace_x3_top; no reference-directory search is required. design/SPEC.md defines the complete public/internal interfaces and ABI, design/ARCHITECTURE.md explains ownership, and tests/README.md indexes verification.

Reset DUT and bridge together and discard outstanding responses. Release reset, select mode while quiescent, configure queues and consume control responses. Populate descriptor, SG and payload storage before posting producer updates; retain it unchanged until completion handshake.

Connect payload, CRC, metadata, control and completion consumers with independent backpressure. Implement bridge alignment, splitting and ordering externally. Legacy cfg_* operations remain distinct from queue configuration.

Establish recipient-specific external input/output delay budgets and physical interface constraints. The passed default-parameter L2 build used virtual pins and a clock-only constraint, so it does not qualify board-level inte rface timing. Reassess timing in the recipient's physical integration. [E043]

From the project root run python3 tests/verify.py --suite behavior, python3 tests/verify.py --suite defects, pyth on3 tests/verify.py --suite synthesis and python3 tests/verify.py --suite delivery. No arguments runs all four. Dependencies are python3, iverilog/vvp, verilator, make, g++ and yosys; hardware gates are external to port able L1.

Suites create fresh evidence with source hashes, tool versions, exact commands and retained results. The acce pted behavioral ledger is evidence/behavior-20261029T160841Z-0e7e8fe4/receipt.json; its matrix rows and runs describe the same executions and must not be double-counted.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addr esses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and regi on capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic roun d-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A1 6 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simu lation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV addr ess scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 16 Verilog/SystemVerilog files; 2339 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/furnace_classifier64.sv	124 total	122 code
rtl/furnace_crc_beat_pipeline.sv	94 total	91 code
rtl/furnace_crc_linear.sv	59 total	55 code
rtl/furnace_crc_output_stage.sv	94 total	90 code
rtl/furnace_crc_packet_accumulator.sv	96 total	92 code
rtl/furnace_crc_tail_pipeline.sv	80 total	77 code
rtl/furnace_descriptor_rings.sv	149 total	149 code
rtl/furnace_dma_reader.sv	107 total	107 code
rtl/furnace_drr_scheduler.sv	114 total	114 code
rtl/furnace_header_parser.sv	412 total	409 code
rtl/furnace_rss_toeplitz.sv	59 total	59 code

Production file	Physical lines	Code lines
rtl/furnace_sg_dma.sv	268 total	267 code
rtl/furnace_token_bucket8.sv	65 total	65 code
rtl/furnace_x1_top.sv	124 total	121 code
rtl/furnace_x2_top.sv	171 total	170 code
rtl/furnace_x3_top.sv	353 total	351 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E009	rtl/furnace_descriptor_rings.sv
E011	rtl/furnace_drr_scheduler.sv
E014	rtl/furnace_sg_dma.sv
E015	rtl/furnace_token_bucket8.sv
E018	rtl/furnace_x3_top.sv
E019	tests/README.md
E020	tests/credits.sv
E021	tests/dma_contract.sv
E022	tests/driver.cpp
E023	tests/ownership.sv
E024	tests/portable_smoke.sv
E026	tests/reference.py
E027	tests/rollover.sv

Reference	Original project-relative record
E028	tests/startup.sv
E029	tests/verify.py
E036	.vulcan/engineering/run-0001/attempt-0005/response.json
E037	.vulcan/engineering/run-0001/attempt-0006/stdout.log
E038	.vulcan/engineering/run-0001/attempt-0007/stdout.log
E039	.vulcan/engineering/run-0001/attempt-0008/stdout.log
E040	.vulcan/engineering/run-0001/attempt-0009/stdout.log
E042	.vulcan/engineering/run-0001/rtl-timing-lint.json
E043	.vulcan/engineering/run-0001/quartus-sta.json
E044	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).